

# Promoting better economics, renewables and CO<sub>2</sub> reduction through trade: A case study for the Eastern Africa Power Pool<sup>☆</sup>

Tom Remy<sup>a</sup>, Debabrata Chattopadhyay<sup>b,\*</sup>

<sup>a</sup> Energy and Extractives Practice, The World Bank, 1818 H St, NW, Washington, DC 20433, USA

<sup>b</sup> Department of Electrical Engineering, University of Queensland, Brisbane, Australia

## ARTICLE INFO

### Article history:

Received 17 December 2019

Revised 24 May 2020

Accepted 27 May 2020

Available online xxxx

### Keywords:

Hydro power

Regional integration

East Africa

Power System Planning

Optimization model

## ABSTRACT

Cross-border power trade in the Eastern Africa Power Pool (EAPP) has been very low for nearly 15 years since its inception in 2005. This analysis uses a least-cost generation and transmission capacity expansion model to assess economic and CO<sub>2</sub> emissions reduction benefits from different levels of integration. These benefits are assessed for several policy scenarios around renewable energy, national energy security and CO<sub>2</sub> emission reduction targets. In addition, we have explored if major drought events may significantly alter the benefits of trade. The World Bank Electricity Planning Model (EPM) is used to represent 11 countries in the EAPP, Eritrea, South Sudan (and SAPP as an external system) for 2020–2030 including all existing, committed and planned generation and interconnectors. Our analysis shows encouraging prospects for EAPP to potentially gain \$7.6 billion from a “Shallow” level of integration wherein countries retain their national plans and interconnections are limited to existing and already committed transmission links. A “Tight” integration that requires generation and (new) interconnection plans to be optimized at a regional level, would increase benefits to \$18.6 billion. Tight integration would be most efficient in meeting a 30% CO<sub>2</sub> emission reduction target by 2030 retaining \$11.8 billion of benefits. This CO<sub>2</sub> target can be met at 3.8 times lower cost of \$6.6 billion compared to the \$25.7 billion needed to meet the target in the Business-as-usual (BAU) scenario of very low trade. We find that imposing a 20% upper limit on imports would still retain 80% of the benefits that allays a large part of the concerns around national energy security. We have also simulated extreme droughts that may reduce benefits of trade due to 20%–30% lower hydro availability from hydro dominated systems like Ethiopia. This indeed affects the benefits of a Shallow integration although more than 60% of base case benefits are still retained. Higher flexibility of a Tight integration, on the other hand, allows for a benefit retention of 89% or more, by adjusting flow volumes and directions of trade. A Tight integration renders a greater ability to the system to reduce the impact of major droughts and meet policy constraints including CO<sub>2</sub> emission and import restrictions at a lower cost, due to the intrinsic flexibility of a stronger and wider network. Since all these issues are very realistic, the additional investments in major cross-border links and institutional challenges should be addressed to achieve significant benefits over the medium term. Overall, our exploration of the key facets of trade in EAPP suggests an overwhelmingly positive case for trade pointing to a rapid increase in trade volume by an order of magnitude over the next decade.

© 2020 International Energy Initiative. Published by Elsevier Inc. All rights reserved.

## Introduction

Cross-border power trade in Africa has received considerable attention for more than two decades (World Bank, 2018a) leading to the formation of power pools including the Southern Africa Power Pool (SAPP) in the mid-nineties followed by the Central, Western and Eastern Africa

Power Pools (CAPP, WAPP and EAPP respectively) over 2003–2005. Volume of trade (as a share of demand) has, however, been low through the life of these power pools. SAPP is the most liquid market in Africa with 3.5% of demand met through trade, while for the EAPP this figure has been below 1% for most years, or 1.5% at best (World Bank, 2018a). There are several barriers to trade in general and cross-border trade in particular, that explain these low numbers in Africa and also in some of the other regions in the world including most parts of Asia and the Middle East. Lack of sufficient political willingness for a variety of reasons including the notions of self-sufficiency and energy security, have been the biggest hurdle to overcome. The creation of EAPP as a regional institution partly overcame the initial hesitation. The lack of a proper framework to identify regional generation and transmission

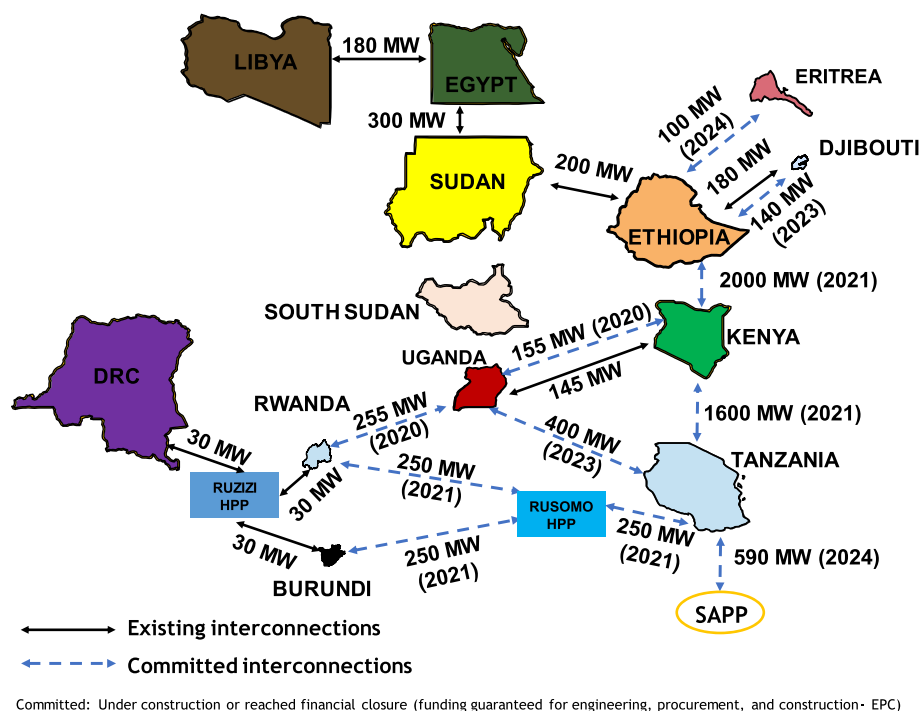
<sup>☆</sup> The authors would like to thank the Editor and an anonymous reviewer for extensive comments that helped them to significantly extend the discussion and clarify the original submission.

\* Corresponding author at: Room I 10-1001, The World Bank, 1818 H St, NW, Washington, DC 20433, USA.

E-mail address: [dchattopadhyay@worldbank.org](mailto:dchattopadhyay@worldbank.org) (D. Chattopadhyay).

There are potentially significant savings from increased trade over the existing interconnectors and a tighter integration by building new interconnections. Integrated operation of a power pool not only reduces reliance on expensive thermal fuel, but also lowers the need for capacity reserve, enhances system reliability and reduces emissions. There is, however, relatively scant literature on regional analyses and estimation of benefits of trade using planning models. The topic has received some attention in the context of SAPP. Bowen et al. (Bowen & Sparrow, 1999),

EAPP's master plans of 2011 (SNC Lavalin & Persons Brinkerhoff, 2011) and its subsequent 2014/15 update (Ea. Energy Analyses



**Fig. 1.** Eastern Africa Power Pool: Existing (black) and committed (blue) interconnections (Ea. Energy Analyses (Energinet.dk), 2014). For committed interconnections, the year in brackets is the year in which the interconnection capacity comes online. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

(Energinet.dk), 2014) showed that 12 new interconnectors can yield discounted total benefits of \$27b over 2013–2038 which is 4.38 times the cost of such developments. A significant part of these benefits stems from EAPP's ability to unleash the hydropotential principally in Ethiopia, but also in DRC, Uganda, and Sudan, to serve the wider region and reduce its heavy reliance on gas. A roadmap for EAPP's development by Deloitte (2015) identified the following key success factors to realize the benefits of increased integration in the EAPP: a common legal and regulatory framework, harmonization of the operational framework (e.g. a common interconnection code), increased power generation capacity, technical and financial support of bilateral and multilateral development agencies (e.g., World Bank, African Development Bank, etc.), the creation of an independent regional regulator, the implementation of a regional power trade market with specific market rules for the EAPP, and the construction of cross-border interconnectors. Oseni and Pollitt (2016) also identified the critical role of these “preconditions” for trading to occur at a large-scale. As we have noted in the preceding discussions, formation of EAPP in itself was an important step to develop an institution that is endowed with the responsibility to grow interconnection and trade. A significant part of the infrastructure is in advanced stage including eight interconnectors that are expected to be commissioned in the next three years. Some of the more capital intensive “hard” preconditions have, therefore, been met, or will be met in the next three years. The “soft” preconditions will need to be acted upon quickly to ensure the challenges that thwarted trade development for 15 years are finally addressed.

The strategic generation and hence transmission choices in the EAPP have, however, not fully been settled in part because of the complexities that arise in the context of large-scale hydro development as well as the variable renewable energy (VRE) revolution that has taken place since the time the master plan (in 2011) was prepared. Sridharan et al. (2019) identified the co-benefits of a strongly interconnected EAPP to “drought-proof” countries like Uganda and Tanzania that are heavily dependent on hydro. Their work demonstrated that the EAPP master plan roll-out would work well in a wetter climate scenario than has been encountered in the past or expected in the future. This will call for a scrutiny of the current interconnection plan to render it more robust. Kammen, Jacome, and Avila (2015) have also criticized the dominance of hydro in the regional master plan and noted “EAPP [should] shift its priorities to include a much greater proportion of renewable energy sources like solar, geothermal and wind, and take greater account of climate risks to large hydropower projects.” The topic of interconnecting hydro-thermal systems for mutual benefits and the associated benefits of specific interconnectors in developing countries generally, has not received as much attention as it should. Cross-border power trade in South Asia has benefitted from some analytical works to establish significant benefits of scale-efficient hydro as well as other renewable developments in the region through a series of studies e.g., (Chattopadhyay, 2013; Wijayatunga, Chattopadhyay, & Fernando, 2015; Timilsina, 2018; South Asia Regional Initiative for Energy Integration (SARI/EI), 2018; South Asia Regional Initiative for Energy Integration (SARI/EI), 2016; Chattopadhyay, 2019).

There is a need for understanding the value proposition of interconnectors in the EAPP that rigorously goes through multiple scenarios of interconnection in an orderly way to understand the relative roles that interconnections, as well as local (solar, wind and geothermal) renewable developments can play to contribute to the least-cost development of the EAPP, while reducing CO<sub>2</sub> emissions from power generation. The present analysis builds on the foundation provided by previous EAPP master plans (SNC Lavalin & Persons Brinkerhoff, 2011; Ea. Energy Analyses (Energinet.dk), 2014) to develop a structured analysis of interconnection development covering a wider range of issues.

Our analysis considers three levels of integration: a Business as Usual (BAU) benchmark, Shallow, and Tight (Fig. 2). The BAU scenario is used as a comparator to assess the benefits of shallow and deep/tight integration.

These three core scenarios represent progressively deeper integration that may achieve greater benefits but would also involve a greater level of coordination for dispatch and system planning that is harder to achieve. The model optimizes dispatch and capacity addition for all scenarios but the domain of optimization changes. In the BAU, trade remains at the current level and the countries independently optimize their generation expansion plans based on country specific demand projections. Dispatch is largely optimized within each country as the volume of trade is minimal in this scenario. Under the ‘Shallow’ integration, each country keeps the same (national) capacity plan as under BAU, but trade is now optimized along the existing and committed interconnectors through coordinated dispatch with trade volume allowed to go up to the transfer capability of these interconnectors (shown in Fig. 1). The model also considers integration with SAPP from 2024. Finally, the ‘Tight’ integration scenario assumes a region-wide optimization of generation and transmission plans. Under ‘Tight’ integration, trade is optimized along existing, committed and candidate interconnections (Fig. 3). Candidate interconnectors are selected endogenously by the model from 2025 if they are economically attractive, i.e., the benefits of the additional interconnection exceed the capital cost of the line. Data for planned interconnections are sourced from existing feasibility studies, or in the absence of any data we have assumed a candidate level of 150 MW of interconnection for the smaller countries.

The study addresses the following set of questions:

1. **What are the benefits of increased integration relative to a Business as Usual (BAU) where trade is restricted to the current level?**
  - a) What level of benefits can be achieved using existing and committed interconnectors through dispatch optimization alone, wherein countries maintain their autonomy of generation planning (i.e., countries make their generation capacity decisions to meet their own national demand)? In this case, each country keeps its optimized least-cost capacity plan as under the BAU scenario, but trade is optimized along existing and committed interconnectors.
  - b) What additional benefits relative to the BAU would accrue from a more holistic optimization of generation as well as new interconnection developments? In other words, the goal is to prepare an optimal capacity plan for the entire EAPP and achieve the maximum

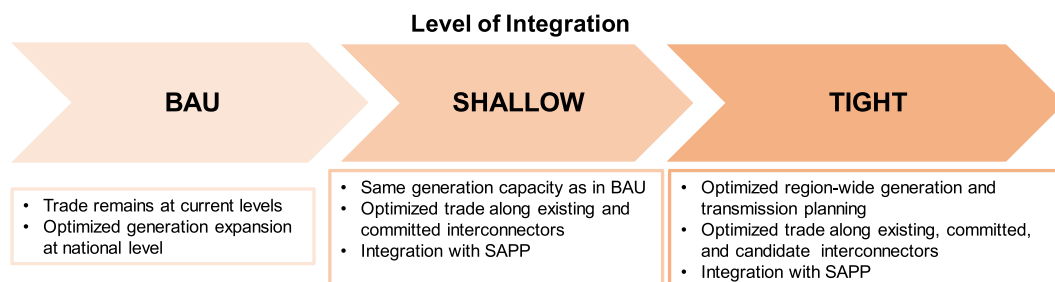


Fig. 2. Different levels of integration under BAU, shallow, and tight integration.

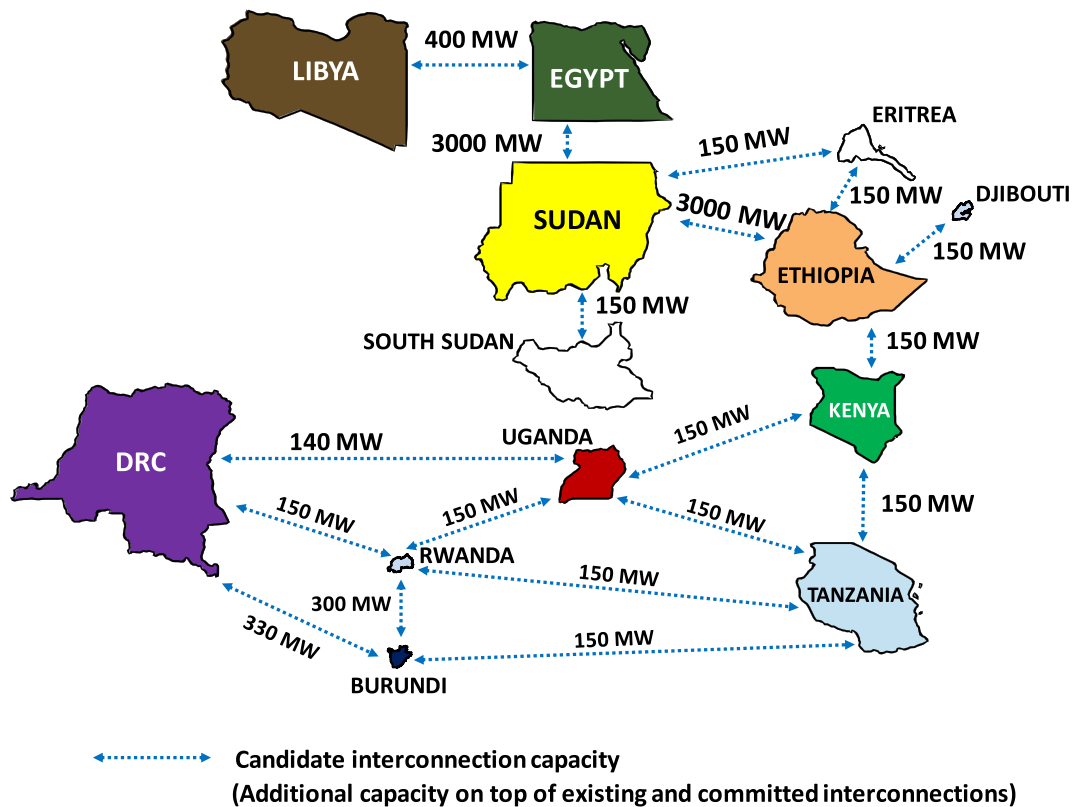


Fig. 3. Eastern Africa Power Pool: candidate interconnections as from 2025. These interconnections are still in planning/conceptual stage.

benefits of trade along existing, committed, and candidate interconnectors.

2. **What are the CO<sub>2</sub> benefits of increased integration?** Would regional trade development help to achieve CO<sub>2</sub> emission reduction at a substantially lower cost?

The first two questions analyze the benefits of trade in terms of costs and climate co-benefits. The following three questions investigate the robustness of the potential gains by assessing the potentially negative impact of renewable energy policies, or supply constraints, on trade.

3. **Do renewable energy targets hamper trade?** Does development of local renewable resources through a country specific RES target diminish the role of trade in the EAPP significantly? Would trade volume and benefits improve significantly if EAPP were to adopt a regional RES target?
4. **Is there a negative effect of national energy security targets?** Do trade benefits drop significantly if countries protect part of the domestic generation through a minimum share of domestic generation in total demand? Is there a threshold for such share below which bulk of the trade benefits are retained?
5. **What is the impact of extreme droughts?** Would trade benefits reduce substantially if the Nile river basin sees extreme droughts?

## Modelling methodology and implementation

### General features of the model

We use the World Bank Electricity Planning Model (EPM) which is fully documented in (Chattopadhyay, 2019; Chattopadhyay, de Sisternes, & Oguah, 2018) and is not reproduced here.

The core mixed integer programming (MIP) multi-zone model, implemented in the General Algebraic Modelling System (GAMS) minimizes the total discounted system costs. The total system costs include annualized investment costs of new generation and transmission projects, fuel costs, fixed and variable O&M, import costs or export revenues of trade with external zones, and the cost of unserved energy. The model optimizes the generation and transmission expansion in the long-term as well as the underlying dispatch of generation and flows on transmission lines for existing and new transmission assets. In addition to the core decision variables on generation and transmission capacity addition, dispatch and flow, the model also co-optimizes spinning reserve provision from the generators. The core EPM accounts for the following constraints relevant to investment and dispatch optimization:

- Demand and capacity reserve requirements: meeting the demand for each load block and meeting the reserve requirement for each load condition;
- Minimum and maximum generation limits for power plants;
- Intermittency of variable renewable resources such as solar and wind on the short, medium and long-term (inter-annual) (Potter Van Loon, Chattopadhyay, & Bazilian, 2020) as well as their impacts on spinning reserve;
- Maximum ramp-up and ramp-down rates of generators;
- Transmission limits between different internal and external zones;
- Joint capacity limit on generators for energy and spinning reserve;
- Spinning reserve requirement constraints;
- Transmission security constraints; and
- Relevant constraints on demand side response.

Some of the additional features of the EPM that have been added to specialized versions of the model are as follows:

1. Modelling a full range of technology options including CSP, thermal and battery storage (Chattopadhyay, Bankuti, Bazilian, de Sisternes, & Sanchez, 2018);
2. Detailed co-optimization of the allocation of spinning reserves (Islam, Khan, Md., Nicolas, & Chattopadhyay, 2019);
3. Recognition of upstream fuel constraints and linkages to coal mine operation and retirement (Nikolakakis, Chattopadhyay, & Bazilian, 2017);
4. Demand side management including demand response and energy efficiency options (World Bank, 2018b);
5. A range of policy options including carbon, renewable and energy efficiency targets (Chattopadhyay, 2019; Chattopadhyay, de Sisternes, & Oguah, 2018);
6. Considering uncertainties in multiple parameters through a stochastic programming variant of the model as well as a Monte Carlo module (Chattopadhyay, 2019; Chattopadhyay, de Sisternes, & Oguah, 2018);
7. EPM can be used as part of Robust Decision Making (RDM) framework to analyze issues around pervasive uncertainty typical of fragile countries (Oguah & Chattopadhyay, 2018; Spyrou, Hobbs, Bazilian, & Chattopadhyay, 2019); and
8. Analysis of resilience of power systems (Mukhi, Chattopadhyay, Hobbs, & Spyrou, 2017).

The model was developed by the Power System Planning Group of the World Bank in 2015 and since then it has been implemented in more than 60 developing countries to inform investment decisions and policy analyses. It is largely deployed for the World Bank's internal analysis and for capacity building exercises for utilities and ministries conducted by the Bank. EPM has also been adopted by some utility planning groups through the World Bank's Technical Assistance projects. The model has also extensively been used for other regional markets such as Central Asia, South Asia, the Middle East, North Africa, Southern Africa and the West Africa Power Pool.

#### *The Electricity Planning Model for the EAPP*

The current implementation of EPM for EAPP analyses the incremental benefits of Shallow and Tight integration relative to the BAU over the period from 2020 to 2030. The economic benefits between the different levels of integration are measured as the differences in total discounted system costs. Cost of unmet demand is set at \$1200/MWh as per the EAPP 2014 Master Plan (Ea. Energy Analyses (Energinet.dk), 2014).

The model represents 11 EAPP countries together with two prospective EAPP members, namely, Eritrea and South Sudan. Barring DRC, each country is modelled as a single zone. DRC is split into four zones: West, South, North, and East. The four zones within DRC cannot trade with each other in all integration scenarios. However, the DRC East zone can trade with the neighboring countries of Rwanda (under Shallow

and Tight integration), Burundi (under Shallow and Tight integration), and Uganda (only under Tight integration). Fig. 1 presents the cross border interconnections for Shallow integration. The model considers SAPP (specifically, Zambia) as an external system represented by historic prices in the 2017 Day Ahead Market (SAPP, 2017) that are used to determine import/export from SAPP at those prices. Trade with SAPP is driven by differences in hourly marginal price between SAPP and the interconnecting zone (Tanzania). Planned/candidate interconnectors shown in Fig. 3 include two major 3 Giga Watt (GW) links connecting Egypt-Ethiopia-Sudan. Candidate interconnectors are co-optimized in EPM, i.e., selection of generation and transmission projects are made simultaneously. This has special significance given that new transmission routes such as Ethiopia-Sudan-Egypt may become available to support the development of large hydro and non-hydro regional renewable projects to serve the entire region better.

EPM analysis is conducted without and with each incumbent candidate interconnector to calculate the benefits in the short-term and long-term (Fig. 4):

The current implementation uses the core EPM (General features of the model section) to model the different integration scenarios (BAU/Shallow/Tight). In addition, we implement the following constraints to capture three policy scenarios:

- Maximum annual CO<sub>2</sub> emission limit at a country level to reach 30% emission reduction by 2030 relative to a Business-as-usual (BAU) scenario;
- Minimum annual renewable target (including hydro) at a country level or alternatively at a regional level expressed as a percentage of domestic electricity requirement; and
- Minimum annual domestic generation share as a proxy for national energy security also as a percentage of domestic electricity requirement.

The modelling framework and its specific implementation for EAPP allow us to squarely address the questions raised in the preceding section. It is important that all of these are discussed in a structured way. We use EPM which renders the flexibility to capture specific policy constraints adds to its value to objectively assess the value of trade in the EAPP.

#### *Key model inputs*

Key input data for the EPM implementation of EAPP include:

- *Load*: load forecasts by country assuming inelastic demand together with planning reserve margins;
- *Generation*: operational and cost characteristics of existing, committed and candidate generation plants;
- *Transmission*: intercountry transmission capacities and transmission

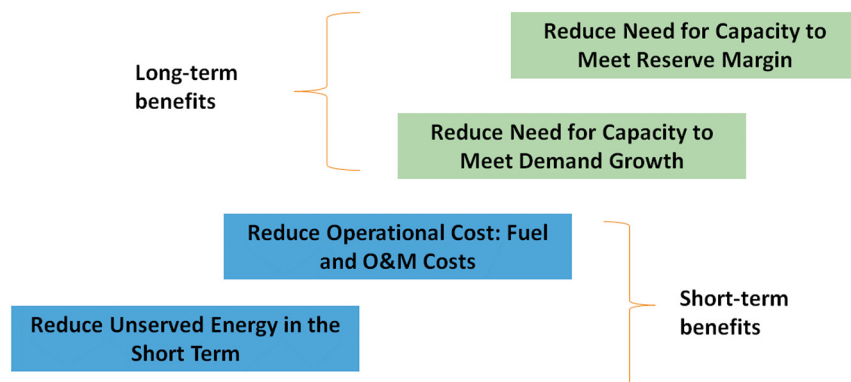


Fig. 4. Short-term and long-term benefits calculated using EPM.

**Table 1**  
Energy requirements (TWh) for EAPP countries, Eritrea and South Sudan.

| Country     | 2020 energy required (TWh) | 2030 energy required (TWh) | CAGR 2020–2030 |
|-------------|----------------------------|----------------------------|----------------|
| Egypt       | 230.8                      | 394.8                      | 5.5%           |
| Sudan       | 24.4                       | 82.1                       | 12.9%          |
| Libya       | 45.6                       | 59.5                       | 2.7%           |
| South Sudan | 1.6                        | 3.9                        | 9.4%           |
| Ethiopia    | 16.3                       | 34.7                       | 7.9%           |
| Kenya       | 12.8                       | 25.8                       | 7.3%           |
| Tanzania    | 13.3                       | 35.7                       | 10.4%          |
| Burundi     | 0.5                        | 0.7                        | 4.7%           |
| Rwanda      | 1.2                        | 3.1                        | 10.0%          |
| Djibouti    | 0.6                        | 1.4                        | 8.1%           |
| Uganda      | 5.1                        | 8.0                        | 4.6%           |
| DRC         | 13.8                       | 23.9                       | 5.6%           |
| Eritrea     | 1.2                        | 2.2                        | 6.1%           |
| Total       | 367.1                      | 675.8                      | 5.2%           |

## CAPEX

### Load

Table 1 shows the energy requirements for all 13 countries. Forecasts for Egypt, Libya, Sudan, South Sudan, DRC, Ethiopia, Tanzania, Djibouti, and Eritrea come from earlier World Bank internal models or external studies conducted for the incumbent utilities. Future energy requirements for Uganda, Rwanda, Burundi, and Kenya are taken from the EAPP 2014 master plan (Ea. Energy Analyses (Energinet.dk), 2014). The model assumes that the total energy requirement is met through grid connected technologies. Egypt is the largest system whose demand (231 TWh pa) accounts for 63% of the total energy requirement in 2020 and its electricity requirements will grow further to 395 TWh in 2030. However, there are much smaller power systems like Sudan (24 TWh pa in 2020) that will grow at a faster rate of 12.9% pa to reach 82 TWh in 2030. Overall energy requirements for EAPP will grow from 367 TWh in 2020 to 676 TWh by 2030.

Yearly demand is modelled using three representative days (in hourly resolution) per quarter i.e., 288 (4 quarters per year \* 3 days per quarter \* 24 h per day) hourly time blocks per year. The first (“peak”) day contains the maximum peak in the quarter. The second (“low”) day has the minimum peak for the quarter. The third (“average”) day contains the average demand per hour in the quarter.

The study assumes a 15% planning reserve margin per country.

### Generation

As existing installed (Fig. 5) and committed (Fig. 6) capacity mixes show, the vast majority of the existing and short-term energy needs in the bigger systems like Egypt and Libya are met through fossil resources, primarily gas and a small share of heavy fuel oil (HFO). Fuel price projections by country are given in Fig. 7.

There is however a much wider array of renewable opportunities up to 172 GW available in the region (Fig. 8) including hydro (Ethiopia, DRC, Tanzania and Uganda in particular), solar (PV and CSP) and wind in Egypt, Libya and Sudan, and geothermal (Ethiopia and Kenya). Fig. 9 shows the portfolio of other candidate projects by country and fuel.

Operational (available capacity, maximum technical availability, maximum energy output of hydroplants per quarter, heat rate, etc.) and cost characteristics [capital expenses (CAPEX), fixed O&M costs (FOM), variable O&M costs (VOM)] of existing, committed and candidate generation plants in Egypt, Libya, Sudan, South Sudan, DRC, Ethiopia, Tanzania, Djibouti, and Eritrea were collected from prior World Bank studies and from the utilities. Power plant data for Kenya were sourced from Kenya’s Power Generation and Transmission Master Plan 2016 (Lahmeyer International, 2016). Characteristics of power plants in Uganda, Rwanda, and Burundi are taken from the EAPP 2014

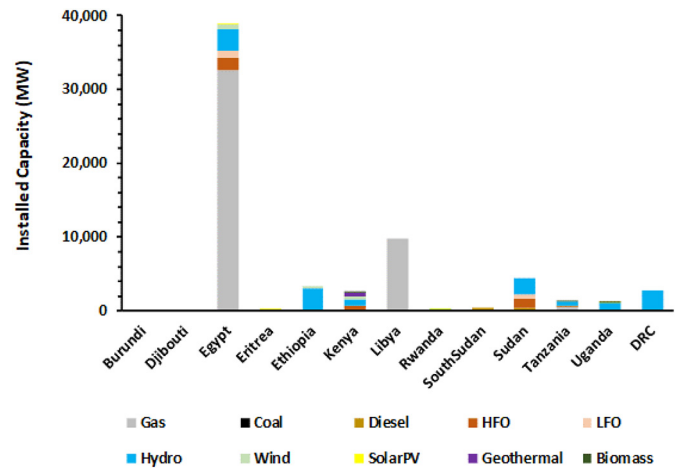


Fig. 5. Installed capacity by country (2020).

Master Plan (Ea. Energy Analyses (Energinet.dk), 2014). Table 2 lists the key operating and cost parameters for candidate plants by country.

**Transmission.** The interconnection capacities of the existing and committed interconnectors is given in Fig. 1. Fig. 3 showed the capacities of potential candidate interconnectors. CAPEX of candidate interconnectors was taken from the EAPP 2014 Master Plan (Ea. Energy Analyses (Energinet.dk), 2014). Transmission CAPEX is annualized over 40 years of economic life of these assets.

### Overview of scenarios

Decisions on generation and associated cross-border interconnection in the EAPP comes down to a choice between thermal vs. renewable in the first instance, and then further as hydro vs. VRE within renewables. The role of trade depends on the generation development path chosen and the level of integration. It would also depend inter alia on policy mandates including renewable, CO<sub>2</sub> emission reduction and minimum domestic generation (or ‘import restriction’) targets. The study uses three core scenarios (BAU/Shallow/Tight) as shown in Fig. 2, to investigate the impact of the level of integration on trade and total costs for the region. Furthermore, we perform five sensitivity analyses across the levels of integration to analyze the impact of specific renewable energy policies (namely, additional availability of RE projects, RE targets, CO<sub>2</sub> emission reduction targets, energy security targets) or supply shocks (namely, a prolonged drought in the region) on the benefits of trade. Table 3 summarizes the different scenarios of the study.

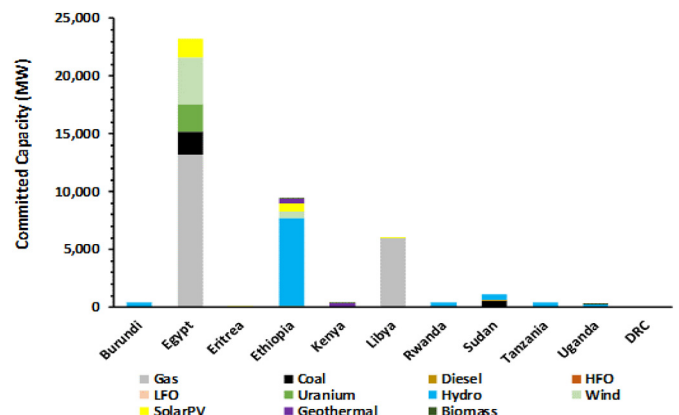


Fig. 6. Committed generation capacity by country for 2020–2030 (MW).

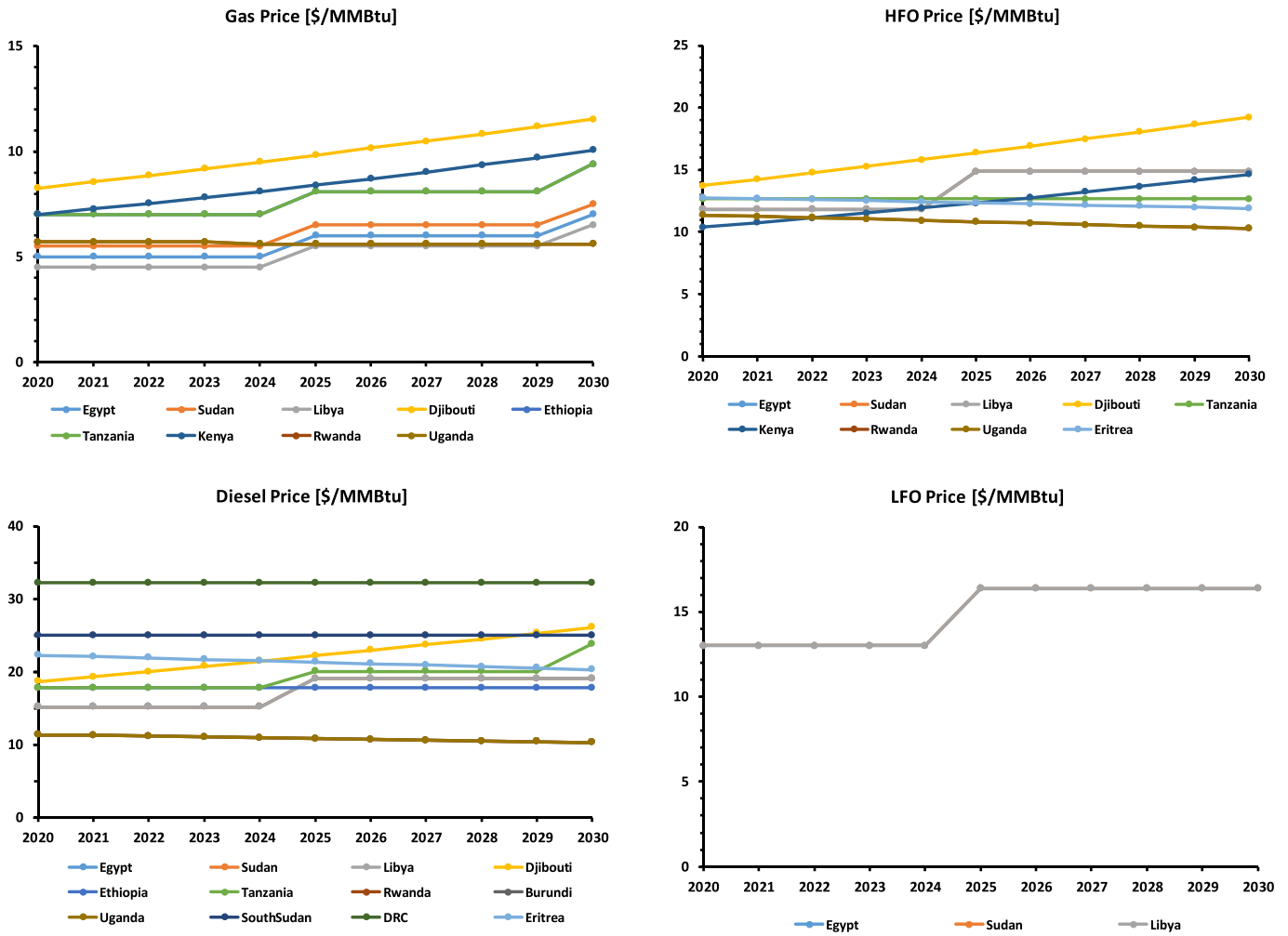


Fig. 7. Fuel price projections for 2020–2030 by country. Note: Projections only include countries that have either existing, committed or candidate plants that use the relevant fuel. LFO price projections are the same for Egypt, Libya and Sudan.

## Results and discussions

**Results: three levels of integration and incremental benefits** section presents the key results for the three core scenarios including the benefits of trade and decomposition of benefits. We analyze the impact of different (renewable) energy policies (additional availability of RE projects, RE targets at the national and regional level, CO<sub>2</sub>

emission reduction targets, and energy security targets) on the benefits of increased regional integration in [Discussions: comparison across main scenarios](#) section. Finally, [Low hydro sensitivity: stress testing trade benefits](#) section describes the results of the drought scenarios and discusses the robustness of the benefits of increased trade in the EAPP region.

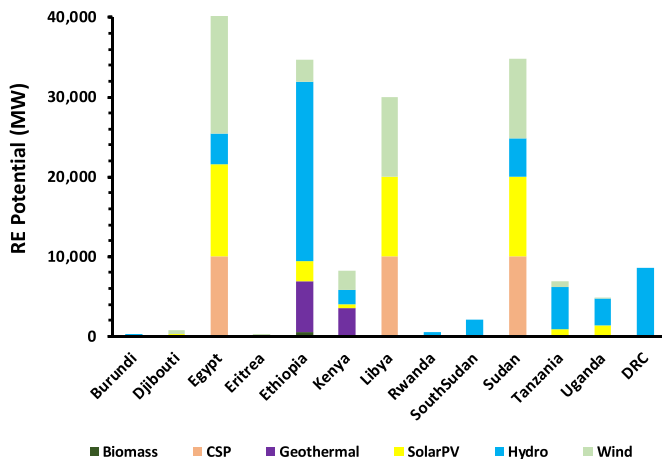


Fig. 8. Renewable energy potential by country (candidate projects).

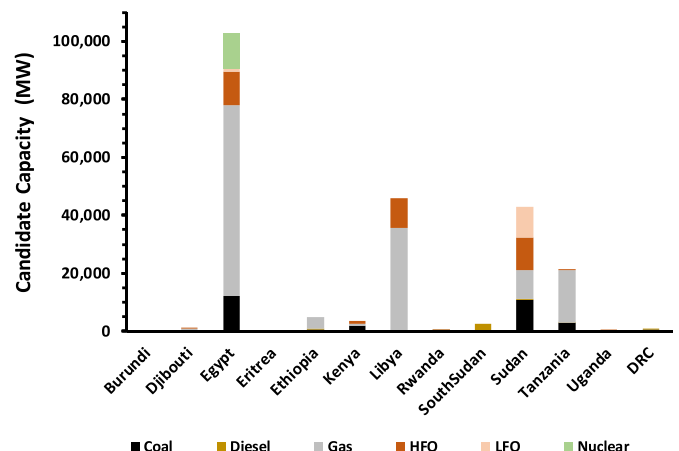


Fig. 9. Fossil and nuclear candidate projects by country.

**Table 2**  
Operating and cost parameters for candidate plants by country<sup>a</sup>.

| Country  | Technology <sup>b</sup> | Fuel       | Candidate available capacity (MW) | Heat rate (MMBtu/MWh) | CAPEX (\$m/MW) | FOM (\$/MW/year) | VOM (\$/MWh) |
|----------|-------------------------|------------|-----------------------------------|-----------------------|----------------|------------------|--------------|
| Egypt    | CCGT                    | Gas        | 10,000                            | 5.8                   | 1.22           | 8370             | 2.24         |
|          | ST                      | HFO        | 10,000                            | 9.4                   | 2.50           | 10,530           | 2.37         |
|          | GT                      | Gas        | 10,000                            | 9.8                   | 0.81           | 8100             | 4.05         |
|          | ST                      | Coal       | 10,000                            | 8.7                   | 2.90           | 32,100           | 4.61         |
|          | Nuclear                 | Uranium    | 10,000                            | 10.5                  | 5.50           | 96,200           | 2.21         |
|          | ROR                     | Water      | 1000                              | –                     | 2.00           | 14,130           | 0            |
|          | Solar PV                | Solar      | 10,000                            | –                     | 0.97           | 10,000           | 0            |
|          | CSP                     | Solar      | 10,000                            | –                     | 4.40           | 40,000           | 4.00         |
|          | Wind turbine            | Wind       | 10,000                            | –                     | 1.49           | 31,000           | 0            |
| Sudan    | CCGT                    | Gas        | 10,000                            | 5.8                   | 1.28           | 8370             | 2.24         |
|          | ST                      | HFO        | 10,000                            | 9.4                   | 2.50           | 10,530           | 2.37         |
|          | GT                      | LFO        | 10,000                            | 9.8                   | 0.81           | 8505             | 4.46         |
|          | ST                      | Coal       | 10,000                            | 8.7                   | 2.90           | 32,100           | 4.61         |
|          | ROR                     | Water      | 2110                              | –                     | 2.00           | 14,130           | 0            |
|          | Solar PV                | Solar      | 10,000                            | –                     | 0.97           | 10,000           | 0            |
|          | CSP                     | Solar      | 10,000                            | –                     | 4.40           | 40,000           | 4.00         |
|          | Wind turbine            | Wind       | 10,000                            | –                     | 1.49           | 31,000           | 0            |
|          | CCGT                    | Gas        | 20,000                            | 5.8                   | 1.25           | 8370             | 2.80         |
| Libya    | ST                      | HFO        | 10,000                            | 9.4                   | 2.50           | 10,530           | 2.37         |
|          | GT                      | Gas        | 10,000                            | 9.8                   | 0.81           | 8100             | 4.05         |
|          | Solar PV                | Solar      | 10,000                            | –                     | 0.97           | 10,000           | 0            |
|          | CSP                     | Solar      | 10,000                            | –                     | 4.40           | 40,000           | 4.00         |
|          | Wind turbine            | Wind       | 10,000                            | –                     | 1.49           | 31,000           | 0            |
|          | CCGT                    | Gas        | 20,000                            | 5.8                   | 1.25           | 8370             | 2.80         |
|          | ST                      | HFO        | 10,000                            | 9.4                   | 2.50           | 10,530           | 2.37         |
|          | GT                      | Gas        | 10,000                            | 9.8                   | 0.81           | 8100             | 4.05         |
|          | Solar PV                | Solar      | 10,000                            | –                     | 0.97           | 10,000           | 0            |
| Ethiopia | CSP                     | Solar      | 10,000                            | –                     | 4.40           | 40,000           | 4.00         |
|          | Wind turbine            | Wind       | 10,000                            | –                     | 1.49           | 31,000           | 0            |
|          | CCGT                    | Gas        | 2000                              | 7.5                   | 1.10           | 24,000           | 2.60         |
|          | ST                      | Biomass    | 496                               | 9.5                   | 4.00           | 44,880           | 3.74         |
|          | ST                      | Coal       | 500                               | 9.5                   | 2.00           | 17,120           | 3.42         |
|          | ICE                     | Diesel     | 140                               | 8.2                   | 1.60           | 60,000           | 6.00         |
|          | ICE                     | Gas        | 2000                              | 10.0                  | 0.70           | 17,400           | 3.50         |
|          | ST                      | Geothermal | 6070                              | –                     | 4.00           | 88,405           | 8.00         |
|          | ROR                     | Water      | 118                               | –                     | 1.48           | 18,000           | 0            |
| Kenya    | Stored hydro            | Water      | 11,770                            | –                     | 2.06           | 18,000           | 0            |
|          | Solar PV                | Solar      | 1800                              | –                     | 0.80           | 21,000           | 0            |
|          | Wind turbine            | Wind       | 1800                              | –                     | 1.90           | 46,000           | 0            |
|          | CCGT                    | Gas        | 1537                              | 6.7                   | 1.03           | 23,172           | 13.40        |
|          | ST                      | Coal       | 1941                              | 9.3                   | 2.11           | 74,560           | 1.35         |
|          | ST                      | Geothermal | 2579                              | –                     | 3.67           | 151,500          | 0            |
|          | ROR                     | Water      | 164                               | –                     | 2.93           | 14,609           | 0.28         |
|          | Stored hydro            | Water      | 930                               | –                     | 3.68           | 17,968           | 0.50         |
|          | Solar PV                | Solar      | 421                               | –                     | 1.70           | 26,000           | 0            |
| Tanzania | Wind turbine            | Wind       | 1891                              | –                     | 2.03           | 76,000           | 0            |
|          | CCGT                    | Gas        | 12,530                            | 9.6                   | 1.20           | 17,120           | 6.28         |
|          | GT                      | Gas        | 18                                | 10.0                  | 0.80           | 17,120           | 3.42         |
|          | ST                      | Coal       | 2900                              | 9.7                   | 2.00           | 17,120           | 3.42         |
|          | ICE                     | Gas        | 5185                              | 9.6                   | 0.80           | 17,120           | 3.42         |
|          | ST                      | Geothermal | 200                               | –                     | 4.36           | 17,120           | 3.42         |
|          | ROR                     | Water      | 4646                              | –                     | 2.49           | 14,700           | 2.62         |
|          | Stored hydro            | Water      | 29                                | –                     | 6.82           | 14,700           | 2.62         |
|          | Solar PV                | Solar      | 700                               | –                     | 1.20           | 23,400           | 0            |
| Uganda   | Wind turbine            | Wind       | 700                               | –                     | 1.57           | 39,700           | 0            |
|          | GT                      | Gas        | 58                                | 9.8                   | 0.81           | 8100             | 4.05         |
|          | ST                      | Coal       | 30                                | 20.1                  | 2.90           | 32,100           | 3.0          |
|          | ROR                     | Water      | 1266                              | –                     | 4.35           | 14,130           | 0            |
|          | Stored hydro            | Water      | 992                               | –                     | 3.34           | 14,130           | 0            |
|          | Solar PV                | Solar      | 1100                              | –                     | 0.97           | 10,000           | 0            |
|          | ICE                     | Coal       | 500                               | 9.8                   | 3.64           | 14,930           | 2.66         |
|          | ROR                     | Water      | 5874                              | –                     | 2.56           | 14,930           | 3.66         |
|          | ROR                     | Water      | 113                               | 1.4                   | 2.59           | 14,450           | 53.15        |
| Burundi  | Solar PV                | Solar      | 8                                 | –                     | 0.97           | 10,000           | 0            |
|          | GT                      | Gas        | 81                                | 8.1                   | 0.81           | 8100             | 4.05         |
|          | ST                      | Biomass    | 53                                | 10.9                  | 2.40           | 44,880           | 4.05         |
|          | ROR                     | Water      | 329                               | –                     | 2.50           | 14,239           | 18.19        |
|          | Solar PV                | Solar      | 30                                | –                     | 0.97           | 10,000           | 0            |
|          | CCGT                    | Gas        | 270                               | 6.6                   | 1.30           | 23,200           | 10.0         |
|          | GT                      | Gas        | 600                               | 7.8                   | 0.80           | 16,000           | 26.0         |
|          | ST                      | Geothermal | 15                                | –                     | 4.00           | 88,400           | 85.0         |
|          | Solar PV                | Solar      | 230                               | –                     | 1.13           | 23,043           | 8.48         |
| Djibouti | Wind turbine            | Wind       | 560                               | –                     | 1.66           | 46,179           | 15.18        |
|          | ST                      | Biomass    | 10                                | 9.4                   | 4.00           | 10,530           | 2.37         |
|          | ICE                     | Diesel     | 5                                 | 8.5                   | 0.43           | 15,000           | 6.00         |
|          | ICE                     | HFO        | 100                               | 6.9                   | 1.60           | 8000             | 4.50         |
|          | Solar PV                | Solar      | 100                               | –                     | 0.97           | 10,000           | 0            |
|          | Wind turbine            | Wind       | 100                               | –                     | 1.49           | 31,000           | 0            |
|          | ICE                     | Diesel     | 1980                              | 7.5                   | 1.50           | 20,081           | 40.00        |
|          | ROR                     | Water      | 2093                              | –                     | 2.21           | 3193             | 5.18         |

<sup>a</sup> Values for heat rate, CAPEX, FOM, and VOM of each technology type are weighted averages over the candidate capacity of that type of technology.

<sup>b</sup> CCGT = combined cycle gas turbine, ST = steam turbine, GT = gas turbine, ROR = run of river, ICE = internal combustion engine.

**Table 3**  
Scenarios developed for the analysis.

| Scenario type        | Key attribute(s)                                                 | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Parameter value                                                                                                                                                                                                                                                                    |
|----------------------|------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Core scenarios       | Level of integration                                             | BAU that limits transfers to historic level, Shallow that retains generation planning autonomy and optimizes trade along existing and committed interconnectors only, and Tight that optimizes generation as well as new interconnections regionally.                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | BAU: ~2 TWh pa or 22 TWh over 2020–30.                                                                                                                                                                                                                                             |
| Sensitivity analyses | Availability of RE projects: base <sup>a</sup> and additional RE | Although hydro (in Ethiopia particularly) received significant attention in former EAPP master plans (SNC Lavalin & Persons Brinkerhoff, 2011; Ea. Energy Analyses (Energinet.dk), 2014), and VRE in other studies (Sridharan et al., 2019; Kammen et al., 2015) – combinations of hydro and VRE in conjunction with regional trade are potentially attractive alternatives. We have therefore considered a “base RE” potential of candidate projects (Fig. 7) that reflects the alternatives considered in (Ea. Energy Analyses (Energinet.dk), 2014) in all scenarios. In addition, we performed a sensitivity analysis (“Additional RE”) where we increase the portfolio of available candidate RE projects with 31.5 GW. | Base RE Potential of candidate projects (172 GW): Fig. 8<br>Additional Potential: 172 GW (Fig. 8) + 31.5 GW<br>The additional 31.5 GW RE potential is split between Egypt: 15 GW (10 GW Solar and 5 GW Wind), Ethiopia: 10 GW (Hydro), and Kenya: 6.5 GW (Geothermal) <sup>b</sup> |
|                      | National and EAPP-wide RE target                                 | In order to put the value of trade in context we also imposed two variants of the RE target – national vs. regional, to check if trade would yield a cheaper solution especially for the regional RE target regime to support cheaper scale-efficient RE projects in the region.                                                                                                                                                                                                                                                                                                                                                                                                                                             | 40% RE generation (as part of the electricity generation mix) target by 2030 at the national or regional level. The starting minimum level of RE generation is set at 20% in 2020.                                                                                                 |
|                      | CO <sub>2</sub> target                                           | National level annual targets in line with the Intended Nationally Determined Contribution (INDC) if the INDC is set in terms of a CO <sub>2</sub> emission reduction target for the power generation sector. However, for many EAPP countries no target has (yet) been set or a target specific for the power sector is not available (UNFCCC, n.d.).                                                                                                                                                                                                                                                                                                                                                                       | 30% CO <sub>2</sub> emission reduction relative to BAU by 2030                                                                                                                                                                                                                     |
|                      | Import restriction as minimum domestic generation share          | Minimum domestic generation or import restriction is set as a minimum threshold on energy requirement that must be met with generators within the country.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Varied from 50% to 95% for all countries                                                                                                                                                                                                                                           |
|                      | Drought                                                          | To assess the impact of a supply shock we simulate a prolonged 5-year drought over the period from 2026 to 2030. During the drought, the energy output of the hydro generators in the region is reduced by 20% or 30%.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 20% to 30% reduction in availability for hydrogenerators from 2026 to 2030                                                                                                                                                                                                         |

<sup>a</sup> Base RE potential of candidate projects is used for all scenarios except additional RE sensitivity.

<sup>b</sup> The values for Ethiopia and Kenya were chosen such that the total RE potential (base + additional) of candidate projects matches the total RE targets for Ethiopia and Kenya as listed in the 2018 study of the Oxford Institute for Energy Studies (2018). For Egypt, the additional RE potential was estimated based on IRENA's 2018 Renewable Energy Outlook for Egypt (IRENA, 2018).

### Results: three levels of integration and incremental benefits

Table 4 lists the key parameters of the three core scenarios around the level of integration. There are clearly significant gains of trade of \$7.6b (in NPV terms over 2020–30 or 3.7% of BAU system cost) under Shallow integration and \$18.6b (or 9.0% of BAU system cost) under Tight integration relative to BAU. The former, in particular, is an encouraging finding because such a significant level of cost reduction can be achieved through ‘soft’ investments in dispatch without additional investments needed in new interconnections (beyond the projects that are already committed). Trade volume grows more than ten-fold in Shallow integration and almost thirty-fold for Tight integration showing enormous potential for increasing trade. The incremental net benefit (i.e., over and above the cost of the candidate interconnectors

in annualized terms over 2025–2030) from Shallow to Tight integration is \$11.0b which is also significant, albeit it would also entail substantial regional coordination of generation and transmission planning (and operation).

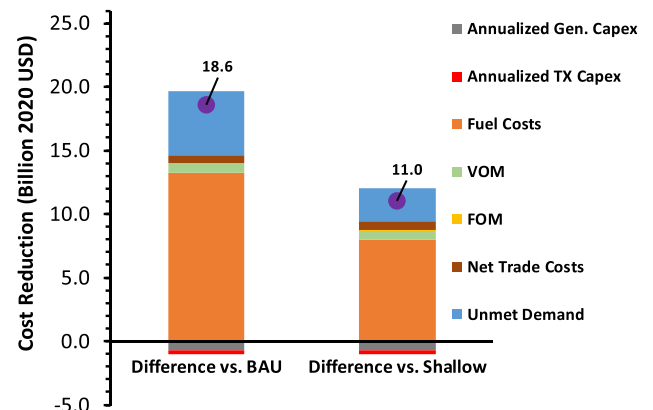
Fig. 10 shows the decomposition of the incremental benefits of Tight integration relative to BAU (\$18.6b) and Shallow (\$11.0b). A move from BAU to Tight can be loosely thought of comprising two phases: (a) a move to Shallow that reduces costs by \$7.6b. Fuel cost reductions of \$5.2b over 11 years (2020–2030) are the largest component of the benefits under Shallow integration. The reduction in fuel costs mainly stems from a \$4.2b reduction in gas costs from \$100.6b under BAU to \$96.4

**Table 4**  
Comparison of key indicators for different levels of integration (cumulative figures for 2020–2030).

| Parameter (2020–2030 total or NPV)                   | BAU         | Shallow integration | Tight integration |
|------------------------------------------------------|-------------|---------------------|-------------------|
| NPV of system costs (\$b) <sup>a</sup>               | 205.7       | 198.1               | 187.1             |
| New capacity (GW)                                    | 99.3        | 99.3                | 101.9             |
| Total generation investments (\$b)                   | 128.8       | 128.8               | 134.9             |
| Interconnection investment (\$m)                     | 0           | 0                   | 348               |
| Total trade (TWh) [% of system demand <sup>b</sup> ] | 21.4 (0.4%) | 229.2 (4.1%)        | 630.7 (11.2%)     |
| Unserved demand (TWh)                                | 38.0        | 35.9                | 32.5              |
| CO <sub>2</sub> emissions (Mt)                       | 1403        | 1363                | 1268              |

<sup>a</sup> Discounted to 2020 at a discount rate of 6%.

<sup>b</sup> System demand in all cases is 5655 TWh [2020–30].



**Fig. 10.** Incremental benefit of Tight Integration relative to BAU and Shallow.

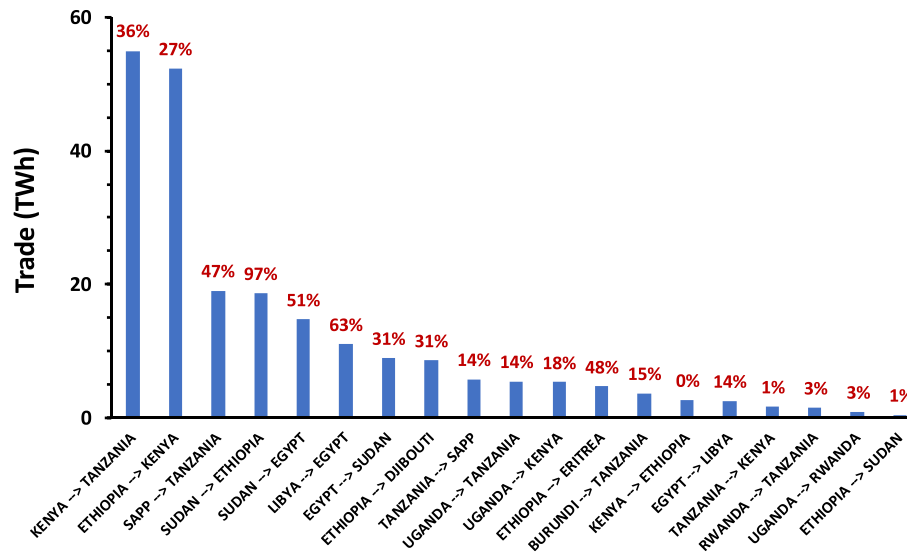


Fig. 11. Volume of trade (TWh) and utilization of links over 2020–2030 (%): SHALLOW.

under Shallow integration. Unserved demand reduction is the second largest component of the benefits and contributes to \$2.4b in benefits; and (b) Shallow to Tight by adding new candidate interconnectors, coordinating generation plans and further increasing trade that primarily saves fossil fuel (mainly gas) generation costs worth \$8.0b. Tight integration also further reduces unmet demand over 2020–2030 by 3.4 TWh with respect to Shallow integration yielding an additional benefit of \$2.6b. The Shallow and Tight states of EAPP are however two distinct *alternatives* that need to be planned differently from the beginning. We discuss this issue further in [Discussions: comparison across main scenarios](#) section.

A residual unserved demand remains even in the Tight integration. Under Tight integration more than 98% of the unserved demand occurs in isolated zones in DRC, demonstrating that Tight integration captures almost all feasible gains from reducing unserved demand in the region. The residual unserved demand indicates two possibilities: (a) not all of projected DRC demand can be met over the next decade; and (b) at least some part of it may be met through off-grid solutions. We do not model the off-grid solutions. Given the primary focus of the work is on benefits

of transmission interconnection that is calculated as the difference among these scenarios, there is no material impact on the residual unserved energy.

Figs. 11 and 12 report the cross-border flows (aggregated over 2020–2030) and average utilization of the links over the next 11 years. Ethiopian export to Kenya/Tanzania, import from SAPP, and export from Sudan to Egypt and Ethiopia supporting growing demand are critical in the Shallow integration scenario. Weak interconnection between Ethiopia and Sudan/Egypt explains the dominant flow direction to southern countries. However, as new interconnection options are presented in the Tight integration scenario, both new 3 GW links Ethiopia → Sudan and Sudan → Egypt (Fig. 6) form part of the optimal investment portfolio and are utilized over 2025–2030. Average utilization of the Ethiopia → Sudan link is 83% over 2025–2030. For Sudan → Egypt the utilization stands at an average of 51% over 2025–2030.

The key difference among the three core integration scenarios is an increase in hydro generation in Ethiopia (Figs. 13 and 15) leading to the displacement of fossil generation in other EAPP countries. Under

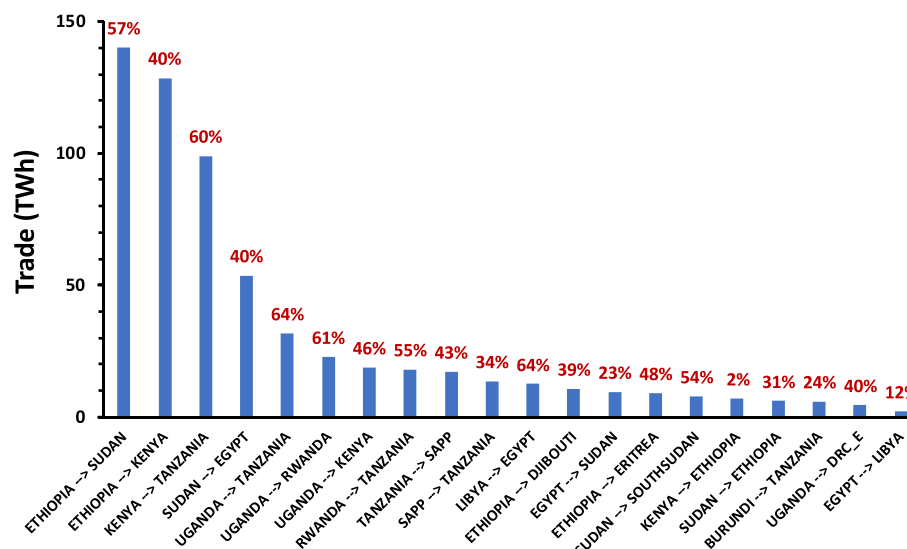


Fig. 12. Volume of trade (TWh) and average utilization of links over 2020–2030 (%): TIGHT.

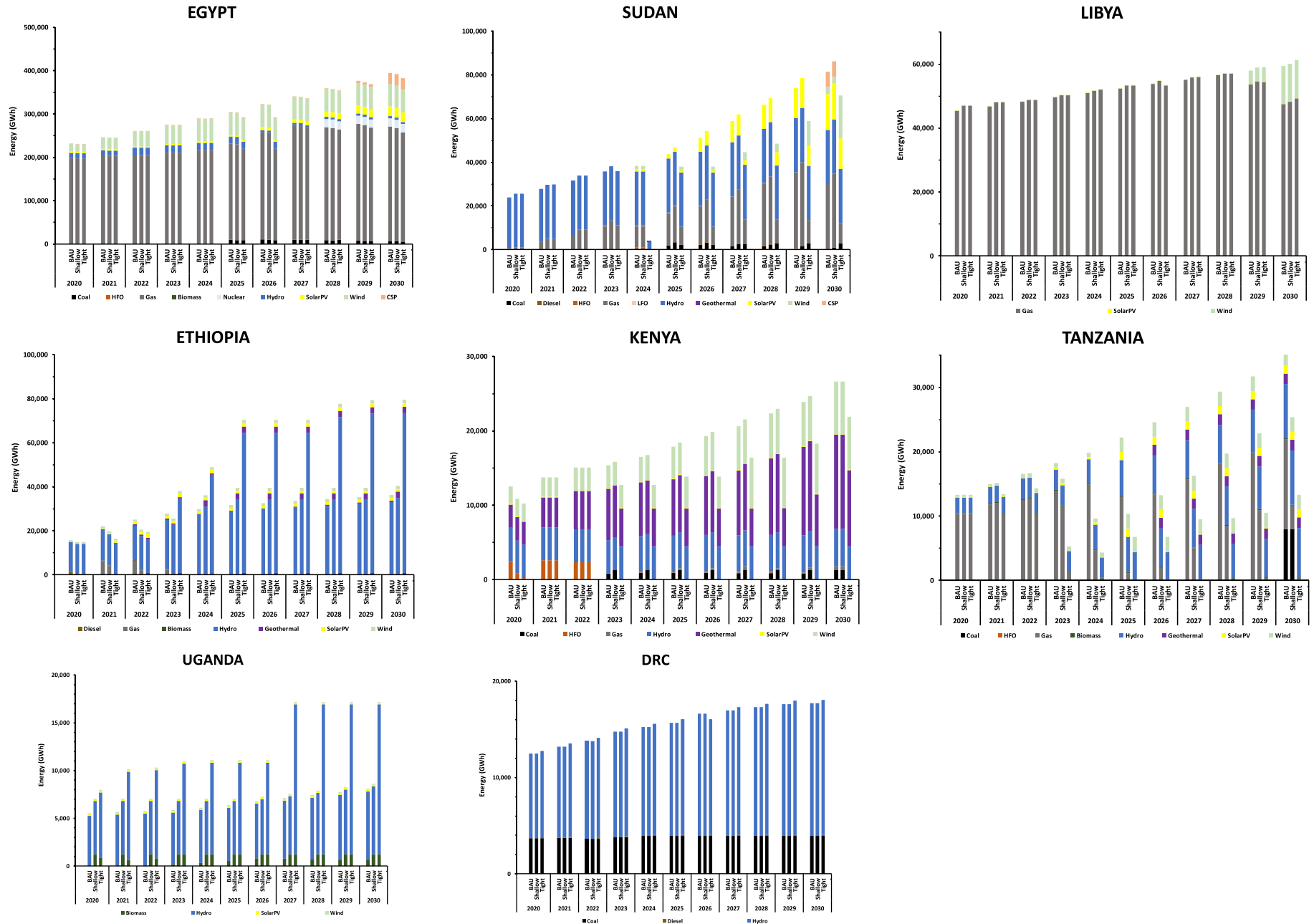


Fig. 13. Electricity generation by fuel over 2020–2030 for top 8 countries in terms of electricity requirements in three core scenarios.

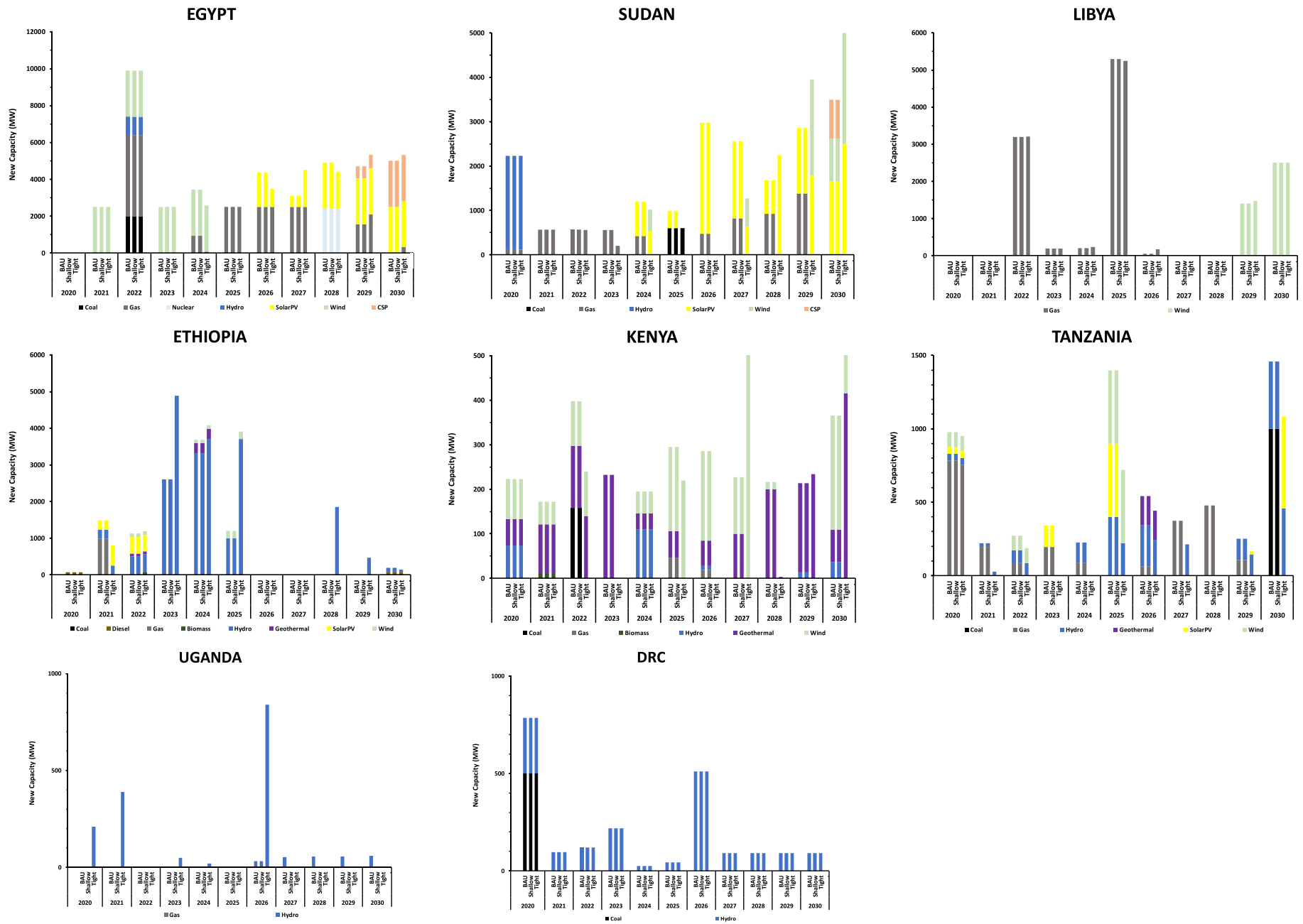


Fig. 14. Capacity expansion plans over 2020–2030 for top 8 countries in terms of electricity requirements in the three core scenarios. \* These 8 countries represent more than 98% of the electricity requirements over the 2020–2030 period.

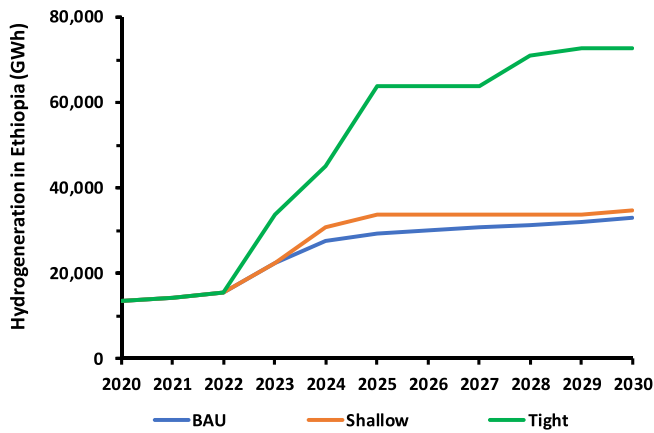


Fig. 15. Ethiopian hydro generation (GWh) for three core scenarios.

Shallow integration, increased hydro generation in Ethiopia from 2024 (Fig. 15) leads to a reduction in gas-based generation in Tanzania (Fig. 16) since the additional Ethiopian power is largely wheeled through Kenya to Tanzania (Fig. 11).

Under Tight integration, the deployment of 7.6 GW additional hydro capacity in Ethiopia (Fig. 14) further displaces thermal generation in Tanzania (Fig. 16), Sudan (Fig. 17) and Egypt as a result of the transfer of power via the new 3 GW links Ethiopia → Sudan and Sudan → Egypt from 2025. There is also a significant increase in Ugandan hydro capacity (Fig. 14) and hydro generation by approximately 40% under Tight integration relative to BAU (Fig. 13), albeit on a much smaller base. In summary, the core scenarios of integration so far demonstrate the value of hydro development in Ethiopia/Uganda to benefit

from regional trade. These findings in many ways reconfirm the general conclusions of the earlier EAPP master plan (SNC Lavalin & Persons Brinkerhoff, 2011; Ea. Energy Analyses (Energinet.dk), 2014). Gas generation elsewhere would fall e.g., gas in Tanzania would practically disappear under Tight integration post 2023 (Fig. 16). The relative shares of gas vs. hydro are as follows: BAU: 63% gas, 17% hydro; Shallow: 61% gas, 18% hydro; Tight: 57% gas, 23% hydro.

#### Discussions: comparison across main scenarios

In this subsection, we discuss the implications of the findings presented in Results: three levels of integration and incremental benefits section. Figs. 17 and 18 show the impacts on total system costs and trade volume for the full array of scenarios (Table 2). Fig. 17 shows how the BAU total system costs of \$207.4b changes across the scenarios. Fig. 17 can be used to derive important insights by comparing across different development paths and policy scenarios. To start with, we have already discussed the BAU → SHALLOW and BAU → TIGHT scenarios that suggest trade can begin using existing and committed lines to achieve \$7.6b in system cost reduction but there is potentially a longer term benefits of \$18.6b from a deep integration of the grids. BAU → SHALLOW and BAU → TIGHT are treated as two alternative development paths for EAPP in our analysis as the direction of arrows in Figs. 17 and 18 shows. Deployment of additional RE resources (Additional RE scenario) can increase the benefits to \$8.8b for Shallow and \$19.3b for Tight suggesting that regional trade and enhanced RE deployment can complement each other well in both integration scenarios.

Trade is even more critical if there is a regional RE target (as opposed to national RE targets). The comparison of national RE targets vis-à-vis an EAPP-wide RES Target shows that the benefits of Shallow integration

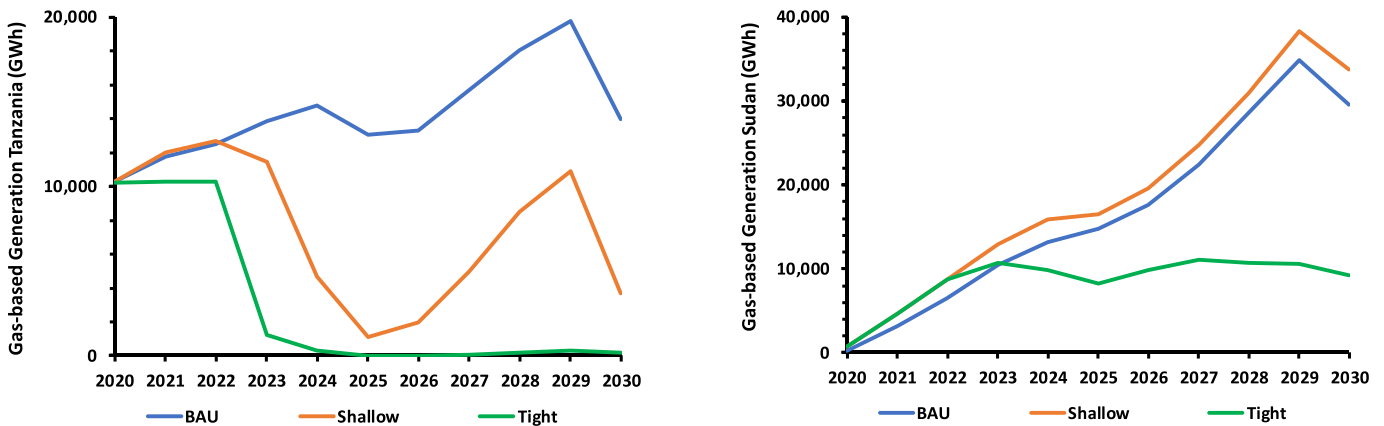


Fig. 16. Gas-based generation (GWh) in Tanzania and Sudan for three core scenarios.

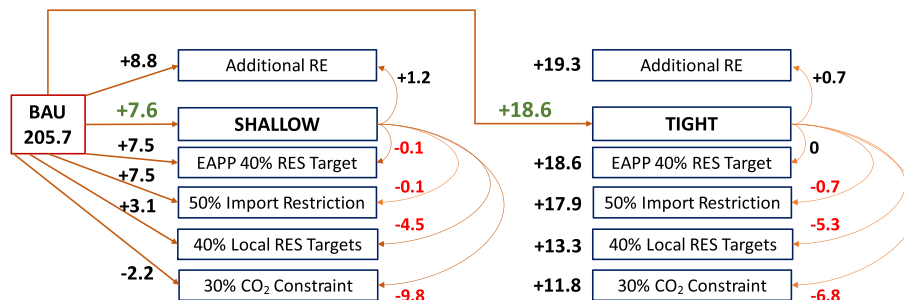
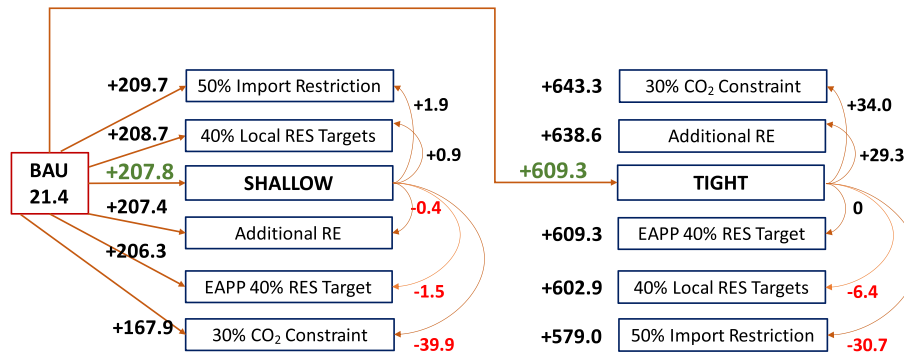


Fig. 17. Incremental benefits in \$b (or costs if negative) across scenarios. Note: Values on the left of each box represent changes (+ refers to benefits) relative to BAU. Values on the right of each box show relative changes vs. the core integration scenarios (Shallow or Tight).



**Fig. 18.** Trade volume (2020–2030) across scenarios (TWh) and relative changes. Note: Values on the left of each box represent changes relative to BAU (+ refers to increase in trade volume). Values on the right of each box show relative changes vs. the core integration scenarios (Shallow or Tight).

would reduce to \$3.1b (from \$7.6b) if national RE targets are imposed. This is because each country develops its own RE portfolio which in turn means greater self-sufficiency and reduced need for (and hence benefit of) trade. On the other hand, an EAPP-wide RE target preserves the benefits of trade under Shallow integration (\$7.6b) even within the limitation of existing/committed lines without altering the total trade volume appreciably. The benefits under Tight integration with an EAPP-wide RE target are also retained and trade volume remains unaffected. Therefore, a greater flexibility in choosing the RE projects to meet a regionwide target can substantially enhance benefits relative to a scenario wherein RE targets imposed at the national level.

A 30% CO<sub>2</sub> emission reduction constraint under Shallow integration reduces the benefits with respect to BAU as the total costs of a 30% CO<sub>2</sub> emission reduction scenario (\$207.9b) are higher than that for the BAU (\$205.7b). However, Shallow integration meets a 30% CO<sub>2</sub> emission reduction target at a cost of \$9.7b, which is significantly lower than the \$25.7b needed to meet the same CO<sub>2</sub> emission reduction constraint under BAU. The CO<sub>2</sub> emission reduction constraint can, in fact, be satisfied at a lower cost of \$6.8b under Tight integration. On the other hand, import restrictions of 50% have a smaller impact of \$0.1b on benefits under Shallow compared to \$0.7b under Tight.

Trade volumes are remarkably close for most of the Shallow scenarios except for a 30% CO<sub>2</sub> emission reduction constraint that sharply reduces the trade flows (Fig. 18). Trade volumes for the Tight scenario nearly triple relative to Shallow in all cases and CO<sub>2</sub> emission constraints in fact increase the volume of trade under Tight.

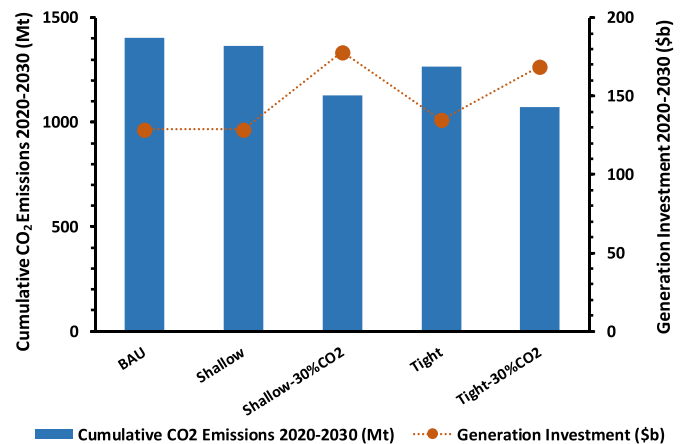
Another set of salient observations from Figs. 17 and 18 is how EAPP could choose between two alternative integration regimes. It is highly beneficial for instance for EAPP to move from BAU → SHALLOW and realize the benefits from the interconnectors under construction through simple dispatch coordination mechanism. It has the advantage of retaining the independence of the member countries to plan for their own generation and no additional transmission investment is needed beyond the committed interconnectors. BAU → TIGHT requires substantial more coordination among planning and concurrent development of large-scale RE projects and additional large interconnectors (e.g. Ethiopia → Sudan) to more than double (net) benefits compared to that of a Shallow integration. If a RE target mechanism is essential to drive RE development, setting an EAPP-wide RES target will help to achieve RE deployment at the lowest possible cost, especially under Tight integration.

A comparison of the BAU → SHALLOW-30%CO<sub>2</sub> and BAU → TIGHT-30%CO<sub>2</sub> transitions further shows that increasing integration (and trade) is a highly cost-effective way to reduce system cost and CO<sub>2</sub> emissions. The Tight-30%CO<sub>2</sub> scenario meets the 30% CO<sub>2</sub> emission reduction target and still achieves an \$11.8b cost reduction relative to the BAU. In the absence of the CO<sub>2</sub> emission reduction constraint, the savings would be \$18.7b. Nevertheless, Tight integration meets the 30% CO<sub>2</sub> emission reduction target at a net cost of \$6.8b which is 3.8 times lower than

the cost to achieve the same CO<sub>2</sub> emission reduction under BAU (\$25.7b). Fig. 19 provides additional details on the effectiveness of the Tight integration regime. Tight-30%CO<sub>2</sub> achieves 55 Mt. less CO<sub>2</sub> emission over 2020–2030 and yet requires \$9.3b lower generation investment compared to the Shallow-30%CO<sub>2</sub> counterpart. We also note that the Tight-30% CO<sub>2</sub> scenario achieves the 30%CO<sub>2</sub> emission reduction at a \$3.0b lower cost than Shallow-30% CO<sub>2</sub> resulting in a benefit of \$14.0b (\$11.0b + \$3.0b) vs. Shallow-30%CO<sub>2</sub> making the Tight-30% CO<sub>2</sub> scenario a highly attractive proposition. In addition, Tight-30%CO<sub>2</sub> increases trade compared to Tight whereas, Shallow-30% CO<sub>2</sub> reduces trade relative to Shallow. A lower cross-border connectivity under Shallow integration forces countries that rely heavily on fossil resources (Egypt, Libya, Tanzania) to deploy local RE resources in early years to reduce CO<sub>2</sub> emissions relative to the BAU. As a result, trade volumes are lower for Shallow-30% CO<sub>2</sub> than for Shallow. The increased interconnectivity under Tight integration allows Egypt to import a larger amount of RE power from Sudan/Ethiopia causing increased trade volumes for Tight-30% CO<sub>2</sub> relative to Tight.

The national energy security constraints expressed as a minimum domestic generation constraint also provides a useful insight. As Fig. 20 shows, system costs barely rise for a minimum domestic generation share of 50%. This suggests a 50% import restriction would still leave most of the benefits intact for both Shallow and Tight integration scenarios. In fact, 80% of the benefits are retained for a 20% import restriction (i.e., 80% minimum domestic generation).

Fig. 21 shows how trade volumes change with Additional RE under Shallow and Tight scenarios. Trade under the Shallow integration scenario is largely unaffected by additional RE, albeit the change in underlying generation mix to include a larger share of



**Fig. 19.** Cumulative CO<sub>2</sub> emissions and generation investment over 2020–2030 under different scenarios.

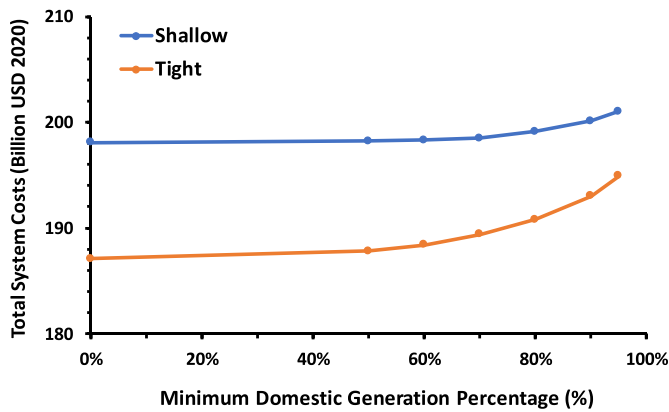


Fig. 20. System cost (\$b) increase with higher domestic generation percentage.

low-cost RE delivers a reduction in system cost of \$1.2b. Tight integration especially boosts the high-volume corridor Ethiopia-Sudan-Egypt with higher penetration of RE (Tight + Additional RE). With the construction of the new 3 GW links Ethiopia → Sudan and Sudan → Egypt, Egypt could deploy additional VRE resources (wind, solar PV) and increase its imports from Sudan/Ethiopia to reduce gas costs and meet increasing demand at lower costs. Since the interconnectivity is limited under Shallow, an increased deployment of RE particularly in Egypt lowers costs but has only limited impact on overall trade.

It should be noted that our analysis suggests hydro as well as solar-wind-geothermal can co-exist and both forms of RE (hydro and non-hydro) can contribute to reduce system costs and reduce the reliance of the system on natural gas. We have, in fact, made a conservative estimate of the total (variable) RE potential of the region mainly based on the EAPP 2014 Master Plan, and will consider more aggressive RE penetration scenarios in our future analysis.

#### Low hydro sensitivity: stress testing trade benefits

Since the benefits of trade are to a large degree linked to hydro developments, these would intricately be linked to the availability of

hydropower plants as highlighted in (Sridharan et al., 2019; Kammen et al., 2015). Lower hydropower generation in Ethiopia/Uganda could reduce volume and value of exports and thus decrease the benefits of trade. On the other hand, a lower hydro availability could also adversely affect the BAU where countries with significant hydropower in the generation mix could see an increase in costs as they need to resort to more expensive thermal power generation resources or suffer increased load shedding. We therefore included a set of sensitivities to essentially stress test the robustness of the gains of trade under a sustained 5-year period of low hydropower availability. Our main objective is to test if low hydrology obviates the need for some hydro capacity in this timeframe and thereby diminishes the benefits substantially. We performed two sensitivities both of which simulate a 5-year drought during 2026–2030 with a 20% (“Hydro80”) and a 30% reduction (“Hydro70”) in available hydro energy for all of the existing and new hydro generators in the Nile river basin. Both these scenarios, and especially the 30% reduction case (Hydro70), represent a relatively extreme dry sequence compared to the Nile river hydrology data presented in the literature (Falchetta, Gernaat, Hunt, & Sterl, 2019; Mulat, Moges, & Moges, 2018).

Table 5 shows the key parameters for the two sensitivities for Shallow and Tight including the relevant benefits that are calculated relative to a BAU which also considers the same dry hydro sequence for 2026–2030. The benefits for Shallow reduce from \$7.6b (core scenario) down to \$5.7b (Hydro80) and further to \$4.8b (Hydro70), representing a maximum loss of benefits of 36%. For the Tight scenario, the reduction in benefits is lower: from \$18.6b in the core scenario down to \$17.1b (Hydro80) and \$16.6b (Hydro70) or a maximum reduction in benefits of 11% (Table 5). As the Tight scenario provides a greater flexibility to re-optimize generation plans at the regional level, a lower reduction in the level of benefits is achieved. Total trade volume over 2020–2030 (552–559 TWh) reduces relative to the core Tight scenario (631 TWh) even though trade increases along different links such as Egypt-Sudan, Sudan-Ethiopia and Kenya-Tanzania (Fig. 22).

The key point to be pondered here is that the added flexibility offered by trade can be very useful even in a highly constrained scenario like a low hydro availability scenario if planning can be done at the regional level.

The flexibility and reversal of trade is particularly relevant for the Tight integration scenario. A comparison of Figs. 12 and 22 shows that as the Ethiopian hydrogeneration decreases Sudan depends more on

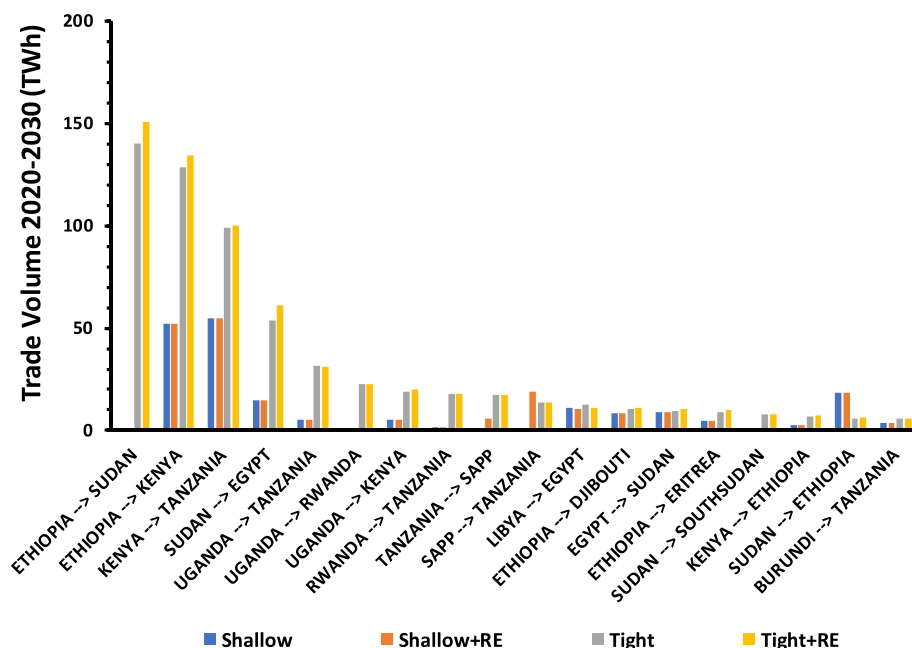
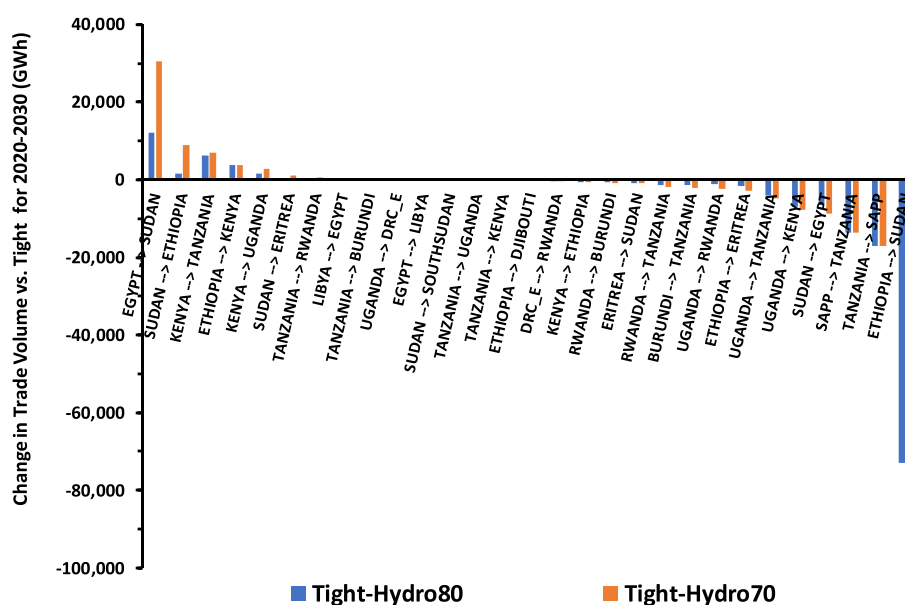


Fig. 21. Comparison of trade volume (2020–2030) across Shallow and Tight with increased RE potential.



**Fig. 22.** Change in trade volume for hydro sensitivity cases under Tight integration.

Egypt rather than on Ethiopia and Egypt could turn into an exporter rather than an importer. Shallow integration retains more than 60% of the base case benefits even in the absence of flexibility of generation planning in an extremely dry scenario.

## Summary and concluding remarks

Regional electricity markets and cross-border power trading have been discussed in the context of all four African power pools (SAPP, EAPP, CAPP, WAPP) over the past two decades. The EAPP was set up in 2005 and its geographical scope has grown significantly over the years. However, the volume of trade has remained extremely low below 1% of total consumption in most years. Although the benefits of trade have been discussed in the past, including two regional master plans prepared in 2011 (SNC Lavalin & Persons Brinkerhoff, 2011) and 2014/15 (Ea. Energy Analyses (Energinet.dk), 2014) – there is value in revisiting the topic in light of the significant hydro as well as non-hydro RE resources that are available in the region. There is also significant increase in cross-border capacity that would take place over the next three years. We have undertaken a comprehensive analysis of the gains of trade taking into account policies and supply scenarios including some of the “downside” scenarios that may potentially reduce trade.

Our analysis reconfirms the previous findings including the most salient aspect around the development of large-scale hydro projects in Ethiopia that can reduce the reliance on gas in some countries like Tanzania. The analysis also provides important insights into how EAPP can initially focus on trade along existing and committed interconnectors

(“Shallow integration”) whereby each country develops its own least-cost generation plan) that can unleash significant benefits over the next five years. Benefits of trade can increase substantially by coordinating generation and transmission planning in the region (“Tight integration”) including major interconnectors with Sudan and Egypt post 2025. Our study compared and contrasted these two alternative levels of increasing integration (Shallow and Tight) against a “Business-as-usual” scenario that restricts trade to the current level till 2030. In addition, we investigated the impact of a select set of generation development and policy scenarios as sensitivity analyses for both the Shallow and Tight integration regimes.

Key findings and resulting policies from our analysis are as follows:

1. Shallow integration alone can save \$7.6b (or 3.7% of BAU system costs) over 2020–2030 in discounted terms. Trade volume needs to grow more than ten-fold from 21 TWh or 0.4% of regional energy requirement under BAU to 229.2 TWh under Shallow or 4.1% of regional energy requirement to achieve these savings. The benefits from Shallow integration accrue from trade along existing and committed interconnectors and thus represent the “low hanging fruits”. Therefore, EAPP and its member utilities need to focus on “soft” infrastructure to reap the benefits of Shallow integration. Such measures should include actions to improve operational readiness (namely, put in place policies and procedures for optimal economic dispatch and creation of central load dispatch centers, prepare a common grid code, etc.) and commercial readiness (namely, agree on trading rules, prepare standardized contracts for power exchange, certify traders, etc.) and provide institutional strengthening within EAPP (namely, capacity building through training, knowledge exchange between member utilities, etc.).
2. Tight integration can significantly deliver significantly higher (net) benefits of to \$18.6b (9.0% of BAU system costs) relative to BAU, but would require substantial new interconnection developments to support additional RE (mainly large reservoir hydro) development. Trade will need to increase as much as 30 times relative to BAU and include major new corridors to support demand in Egypt and Libya through Ethiopian hydro as well as other renewables. Integrated planning at the regional level will need to identify critical ‘regional’ generation and transmission developments needed in the longer term over 2025–2030. The update of the EAPP 2014 Master

**Table 5**  
Low hydro sensitivity results<sup>a</sup>.

|                                | Shallow<br>Hydro80 | Shallow<br>Hydro70 | Tight<br>Hydro80 | Tight<br>Hydro70 |
|--------------------------------|--------------------|--------------------|------------------|------------------|
| Benefit (\$b)                  | 5.7                | 4.8                | 17.1             | 16.6             |
| Trade (TWh)                    | 168.1              | 173.5              | 558.8            | 551.8            |
| Unserved demand (TWh)          | 36.4               | 36.6               | 32.5             | 32.5             |
| CO <sub>2</sub> emissions (Mt) | 1405               | 1427               | 1280             | 1295             |

<sup>a</sup> Benefits are measured vs. the corresponding BAU scenario (BAU-Hydro80 or BAU-Hydro70 respectively).

Plan is a first essential step towards Tight integration with the prioritization of key interconnectors.

3. Increased integration can help to reduce the reliance on gas in the region from 63% (of the electricity mix over 2020–2030) in BAU to 58% in Tight through the development of hydro and non-hydro RE resources and result in 10% reduction in CO<sub>2</sub> emissions or 135 million tons of CO<sub>2</sub> over 2020–2030. These climate co-benefits further build the case for increased integration within the EAPP. Tight integration of EAPP would be a very effective strategy to meet a 30% CO<sub>2</sub> emission reduction target by 2030. A 30% CO<sub>2</sub> emission reduction relative to BAU can be accommodated at a cost of \$6.8b compared to \$9.8b needed under Shallow integration, and \$25.7b under BAU.
4. Imposing a national level RE target (40% of generation mix by 2030) can wipe \$5.3b of the benefits (from \$18.6b down to \$13.3b) for Tight integration because each country becomes more self-reliant. However, if the RE target is set at a regional level, 100% of the benefits is retained. EAPP should ideally consider a regionwide RE target to achieve cost-efficient deployment of (variable) RE resources in the region.
5. A second factor that can diminish the benefits of trade is a national energy security policy enforcing a minimum share of national demand that must be met with local generation resources. Our analysis shows that almost 80% of the benefits of regional trade can still be retained both under Shallow and Tight integration, even if an 80% minimum share of domestic generation constraint is imposed.
6. Finally, a third factor that may reduce benefits of trade are droughts that would reduce flows from hydro dominated systems like Ethiopia and Uganda. This indeed affects the benefits of a Shallow integration although more than 60% of base case benefits are retained even under an extreme drought scenario. Higher flexibility of Tight integration allows one to retain at least 89% of the benefits by adjusting the trade volumes and flow directions.

In summary, we find a resounding economic case for regional trade and a plausible pathway to reap more than 40% of the long-term benefits through dispatch and trade optimization over existing and committed cross-border links before scaling up additional (RE) generation and new cross-border links. We also find that a Tight integration renders potentially better ability to the system to deal with major droughts, albeit this needs a more thorough examination of the operational issues. A Tight integration would also entail a greater ability to meet policy constraints including CO<sub>2</sub> emission and import restrictions due to the intrinsic flexibility of a stronger and wider network. Since all these issues are very realistic, the additional investments in major cross-border electricity trade links and institutional challenges are worth considering in light of such significant benefits over the medium term. Put differently, the major planned interconnectors like Ethiopia-Sudan-Egypt have an “option value” in drought-proofing the region and add to the resilience of the regional power system, that must be brought into the equation when evaluating these investments. Finally, development of scale-efficient regional generation and transmission projects would support extension of electricity access in the region. As the recently proposed Tanzania-Zambia interconnection (TAZA) project by the World Bank (2018c) has argued, cross-border interconnection projects can improve access to currently unserved or underserved customers. For instance, it is estimated that TAZA alone can enable connecting 21,000 new customers in Tanzania. As EAPP develops, it is expected that development of an integrated power system in the region would also bolster the electricity access and increase the pace of connecting new customers rapidly.

### Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

### References

- Bowen, B., & Sparrow, F. T. (1999). Modeling electricity trade policy for the twelve nations of the Southern African Power Pool (SAPP). *Utilities Policy*, 183–197. [https://doi.org/10.1016/S0957-1787\(99\)00019-3](https://doi.org/10.1016/S0957-1787(99)00019-3).
- Chattopadhyay, D. (2013). Cross-border power trading in South Asia: Modelling analysis to assess economic benefits. *2013 IEEE Power & Energy Society General Meeting (PESGM)*, Vancouver, BC, Canada. <https://doi.org/10.1109/PESGM.2013.6672071>.
- Chattopadhyay, D. (2019). *Towards a South Asian regional electricity market: Assessment of economic benefits*. (Presented at the South Asian Power Secretaries Roundtable, Colombo, Sri Lanka).
- Chattopadhyay, D., Bankuti, M., Bazilian, M. D., de Sisternes, F., & Sanchez, M. J. M. (2018). Capacity planning model with CSP and battery. *2018 IEEE Power & Energy Society General Meeting (PESGM)*. Portland, OR, USA. <https://doi.org/10.1109/PESGM.2018.8586160>.
- Chattopadhyay, D., de Sisternes, F., & Oguah, S. (2018). *World Bank Electricity Planning Model (EPM): Mathematical formulation*. (Washington DC).
- Deloitte (2015). *The roadmap to a fully integrated and operational East African Power Pool*. Ea. Energy Analyses (Energinet.dk) (2014). *EAPP regional power system master plan*. (Addis Ababa).
- EAPP (2013). A regional organization adopted as a specialized institution of COMESA (2013). Retrieved from <http://eappool.org/>.
- Falchetta, G., Gernaat, D. E. H. J., Hunt, J., & Sterl, S. (2019). Hydropower dependency and climate change in sub-Saharan Africa: a nexus framework and evidence-based review. *Journal of Cleaner Production*, 231, 1399–1417. <https://doi.org/10.1016/j.jclepro.2019.05.263>.
- IRENA (2018). *Renewable energy outlook – Egypt*.
- Islam, M. E., Khan, Md., Monower Zahid, Nicolas, C., & Chattopadhyay, D. (2019). Planning for direct load control and energy efficiency: A case study for Bangladesh. *IEEE Power & Energy Society General Meeting (PESGM)*. Atlanta, GA, USA. <https://doi.org/10.1109/PESGM40551.2019.8973505>.
- Kammen, D. M., Jacome, V., & Avila, N. A. (2015). *A clean energy vision for East Africa: Planning for sustainability, reducing climate risks and increasing energy access*.
- Lahmeyer International (2016). *Development of a power generation and transmission master plan, Kenya*.
- Mukhi, N., Chattopadhyay, D., Hobbs, B., & Spyrou, E. (2017). *Building climate resilience into power system planning: The case of Bangladesh*. Working Paper ACS23320 Washington DC: World Bank.
- Mulat, A. G., Moges, S. A., & Moges, M. A. (2018). Evaluation of multi-storage hydropower development in the upper Blue Nile River (Ethiopia): regional perspective. *Journal of Hydrology*, 16, 1–14. <https://doi.org/10.1016/j.jhr.2018.02.006>.
- Nikolakakis, T., Chattopadhyay, D., & Bazilian, M. (2017). A review of renewable investment and power system operational issues in Bangladesh. *Renewable and Sustainable Energy Reviews*, 68, 650–658. <https://doi.org/10.1016/j.rser.2016.10.016>.
- Oguah, S., & Chattopadhyay, D. (2018). Planning in fragile and conflict states: Case study for West Bank and Gaza. *IEEE Power & Energy Society General Meeting (PESGM)*. Portland, OR, USA. <https://doi.org/10.1109/PESGM.2018.8586497>.
- Oseni, M. O., & Pollitt, M. G. (2016). The promotion of regional integration of electricity markets: Lessons for developing countries. *Energy Policy*, 88, 628–638. <https://doi.org/10.1016/j.enpol.2015.09.007>.
- Oxford Institute for Energy Studies (2018). *The politics of renewable energy in East Africa*.
- Potter Van Loon, A., Chattopadhyay, D., & Bazilian, M. (2020). Atypical variability in TMY-based power system plans. *Energy for Sustainable Development*, 54. <https://doi.org/10.1016/j.esd.2019.09.004>.
- SAPP (2017). Retrieved from [www.SAPPmarket.com](http://www.SAPPmarket.com).
- SNC Lavalin & Persons Brinkerhoff (2011). *Regional power system master plan and grid code study*.
- South Asia Regional Initiative for Energy Integration (SARI/EI) (2016). *Impact of cross-border power trade on the economy of Bhutan*. (Working Paper IRADe-SARI/EI-WP-2016-01).
- South Asia Regional Initiative for Energy Integration (SARI/EI) (2018). *Gains from multilateral electricity trade among BBIN country: Modelling study*. (prepared by Integrated Research and Action for Development (IRADe), New Delhi, India).
- Spyrou, E., Hobbs, B. F., Bazilian, M., & Chattopadhyay, D. (2019). Planning power systems in fragile and conflict-affected states. *Nature Energy*, 4, 300–310. <https://doi.org/10.1038/s41560-019-0346-x>.
- Sridharan, V., Broad, O., Shivakumar, A., Howells, M., Boehlert, B., Groves, D. G., ... Cervigni, R. (2019). Resilience of the Eastern African electricity sector to climate driven changes in hydropower generation. *Nature Communications*, 10. <https://doi.org/10.1038/s41467-018-08275-7>.
- Timilsina, G. R. (2018). *How would cross-border electricity trade stimulate Hydropower development in South Asia?* (World Bank Policy Research Working Paper 8513).
- UNFCCC (d). Indicative nationally determined contribution. Retrieved from <https://www4.unfccc.int/sites/submissions/INDC/Submission%20Pages/submissions.aspx> (Accessed 25 Nov 2019).
- Wijayatunga, P., Chattopadhyay, D., & Fernando, P. N. (2015, August). *Cross-border power trading in South Asia: A techno economic rationale*. ADB South Asia. (Working Paper 38).
- World Bank (2018a). *Power Trade in Africa*, Washington DC.
- World Bank (2018b). *Poland energy transition – The path to sustainability in the electricity and heating sector*.
- World Bank (2018c). Tanzania Zambia interconnection project, project appraisal document, report no: PAD2516. Retrieved from <http://documents.worldbank.org/curated/en/325581529551846176/text/TANZANIA-PADNEW-05302018.txt>.
- World Bank (2019, June). *Southern Africa Power Pool – Program for accelerating transformational energy projects*, Washington DC.